

=====

DOCKER

=====

TASK1 - DOCKER INSTALLATION

ls /etc/yum.repos.d[root@master ~]#

OR

```
yum repolist
yum install -y docker-ce
ip add | less
```

Note - check for the interface name docker0, it is not available

```
yum list installed | grep docker
systemctl start docker.service
ip add | less
```

Note - check for the interface name docker0, it is now available

```
systemctl enable docker.service
systemctl status docker.service
```

```
docker --help | less
docker ps --help
docker version
docker system info | less
ls /var/lib/docker
```

TASK 02 - DOCKER BASIC COMMANDS

docker images -a

Note

- Open second tab in the same terminal window and execute "docker ps"
- press Ctrl + c in first tab

```
docker ps
docker ps -a
```

```
docker start <container-id>/<container-name>
docker ps
docker stop <container-id>/<container-name>
docker ps -a
docker start <container-id>/<container-name>
docker ps
docker pause <container-id>/<container-name>
docker ps
docker unpause <container-id>/<container-name>
```

```
docker ps -a
```

```
ls
cd containers
ls
cd <container-id>
ls
cat hostname
less <container-id>-.json.log
cd
```

```
docker create quay.io/gauravkumar9130/nginxdemo
docker ps
docker ps -a
```

```
docker images
cd /var/lib/docker/image/overlay2/imagedb/content/sha256/
docker ps -a
docker rm <container-id>
docker ps -a
ls /var/lib/docker/containers
docker rm -f <container-id>
docker ps -a
ls /var/lib/docker/containers
```

```
docker run -d quay.io/gauravkumar9130/nginxdemo
```

```
docker rm -f <container-id>
docker ps -a
docker images
```

```
docker images
ls /var/lib/docker/image/overlay2/imagedb/content/sha256/
```

```
docker pull quay.io/gauravkumar9130/nginxdemo
docker images
ls /var/lib/docker/image/overlay2/imagedb/content/sha256/
```

```
docker run quay.io/gauravkumar9130/ubuntu-git
```

```
docker images
docker ps -a
docker run quay.io/gauravkumar9130/ubuntu-git sleep 120
```

Note

- Open second tab in the same terminal window and execute "docker ps"
- Return back to first tab and execute "docker ps -a"

```
docker run -it quay.io/gauravkumar9130/ubuntu-git
pwd
exit
docker ps
docker ps -a
docker run -i quay.io/gauravkumar9130/ubuntu-git
```

```
pwd
ls /
exit
docker run -t quay.io/gauravkumar9130/ubuntu-git
pwd
ls /
Note - execute below commands in second tab
docker ps
docker stop <cont-id>
Note - Return back to first tab and see the outcome
```

```
yum install -y docker-ce
```

```
docker ps -qa
docker rm -f $(docker ps -qa)
docker ps -a
docker run -itd --name=webapp quay.io/gauravkumar9130/ubuntu-git
docker ps
```

```
docker rename webapp webapp2
docker ps
docker exec <cont-id> hostname
docker exec webapp ps -ef
docker exec -it webapp /bin/bash
ps -ef
exit
docker ps
```

```
docker inspect webapp | less
docker stats
docker top webapp2
docker system events --since 60m
```

```
docker ps
docker kill webapp2
docker rm webapp2
docker run -it --rm quay.io/gauravkumar9130/ubuntu-git
ls
exit
docker ps -a
```

```
docker run -itd --restart always quay.io/gauravkumar9130/ubuntu-git
systemctl restart docker.service
docker ps
```

```
docker run -itd --restart on-failure quay.io/gauravkumar9130/ubuntu-git
docker exec <container-id>/<container-name> ping
docker ps
```

=== port binding/publishing ===

Note - Open second tab [TAB2] in the same terminal window and execute the below

```
ssh root@node1
ip add | less
```

```
ping -c2<master-IP>
```

Note - Return back to first tab [TAB] of the terminal and execute the below
docker run -d -p 127.0.0.1:8080:80 nginx

```
curl localhost:8080
```

TAB2

```
curl localhost:8080
```

TAB1

```
docker rm -f <container-id>
```

-

```
=====
```

TAB1

```
docker run -d -p 8080:80 nginx
docker inspect <container-id> | grep IPAddress
curl <container-IP>:80
```

TAB2

```
curl <master-IP>:8080
```

```
docker system df
docker system df -v
```

```
=== docker image layer\persistence demo ===
```

```
docker run -it quay.io/gauravkumar9130/ubuntu-git
cp /etc/passwd/tmp/mypasswd
ls /tmp
cat /tmp/mypasswd
exit
docker ps -a
docker start <cont-id>
docker ps
docker exec <cont-id> cat /tmp/mypasswd
docker rm -f<cont-id>
```

Image

```
cd /var/lib/docker/volumes
docker volume ls
docker volume create my-volume
docker volume inspect my-volume
docker run -it -vmy-volume:/test --name app1 quay.io/gauravkumar9130/ubuntu-git
cd test
echo "hello"> hello.txt
ls
exit
cat /var/lib/docker/volumes/my-volume/_data/hello.txt
```

```
docker run -it -v my-volume:/test2 --name app2 quay.io/gauravkumar9130/ubuntu-git
cd test2
ls
exit
docker rm -f app1 app2
```

```
docker network ls
docker network create -d bridge mybr
docker network ls
docker network inspect mybr | less
docker run -d -p 80:80 --name app --network=mybr quay.io/gauravkumar9130/nginxdemo
docker inspect app | grep IPAddress
```

```
<in other terminal>
docker run -it --name client1 quay.io/gauravkumar9130/ubuntu-git
apt-get install curl
curl 172.18.0.2 --connect-timeout 5
ip add
exit
docker run -it --name client2 --network=mybr quay.io/gauravkumar9130/ubuntu-git
ip add
docker run -it --name client2 --network=mybr quay.io/gauravkumar9130/ubuntu-git
```

```
yum install -y docker-ce
```

```
yum install -y docker-ce
c
```

```
exit
docker ps -a
docker rm -f $(docker ps -qa)
```

<https://docs.docker.com/engine/storage/drivers/images/container-layers.webp>